Feature Reduction and Selection

Introduction

- In practical multicategory applications, it is not unusual to encounter problems involving tens or hundreds of features.
- Intuitively, it may seem that each feature is useful for at least some of the discriminations.
- In general, if the performance obtained with a given set of features is inadequate, it is natural to consider adding new features.
- Even though increasing the number of features increases the complexity of the classifier, it may be acceptable for an improved performance.

Introduction

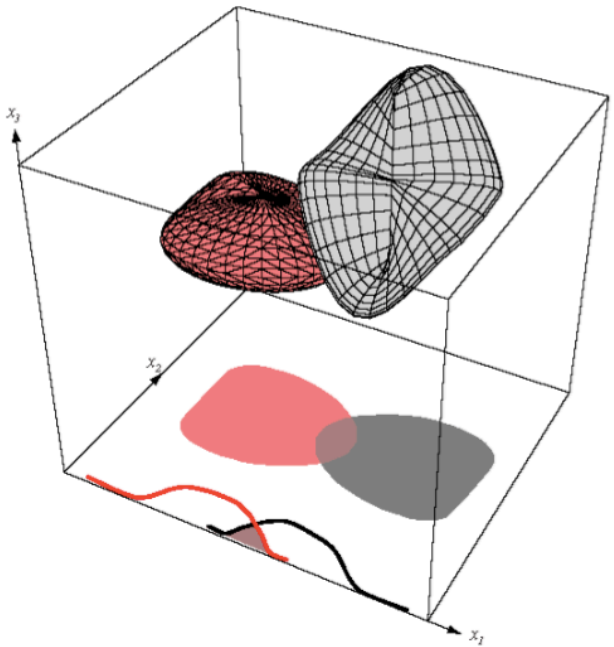


Figure 1: There is a non-zero Bayes error in the one-dimensional x_1 space or the two-dimensional x_1, x_2 space. However, the Bayes error vanishes in the x_1, x_2, x_3 space because of non-overlapping densities.

Problems of Dimensionality

- Unfortunately, it has frequently been observed in practice that, beyond a certain point, adding new features leads to worse rather than better performance.
- This is called the *curse of dimensionality*.
- There are two issues that we must be careful about:
  - How is the classification accuracy affected by the dimensionality (relative to the amount of training data)?
  - How is the complexity of the classifier affected by the dimensionality?

Problems of Dimensionality

- Potential reasons for increase in error include
  - wrong assumptions in model selection,
  - estimation errors due to the finite number of training samples for high-dimensional observations (overfitting).
- Potential solutions include
  - reducing the dimensionality,
  - simplifying the estimation.

Problems of Dimensionality

- Dimensionality can be reduced by
  - redesigning the features,
  - selecting an appropriate subset among the existing features,
  - combining existing features.
- Estimation errors can be simplified by
  - assuming equal covariance for all classes (for the Gaussian case),
  - using regularization,
  - using prior information and a Bayes estimate,
  - using heuristics such as conditional independence,
  - ....

Problems of Dimensionality

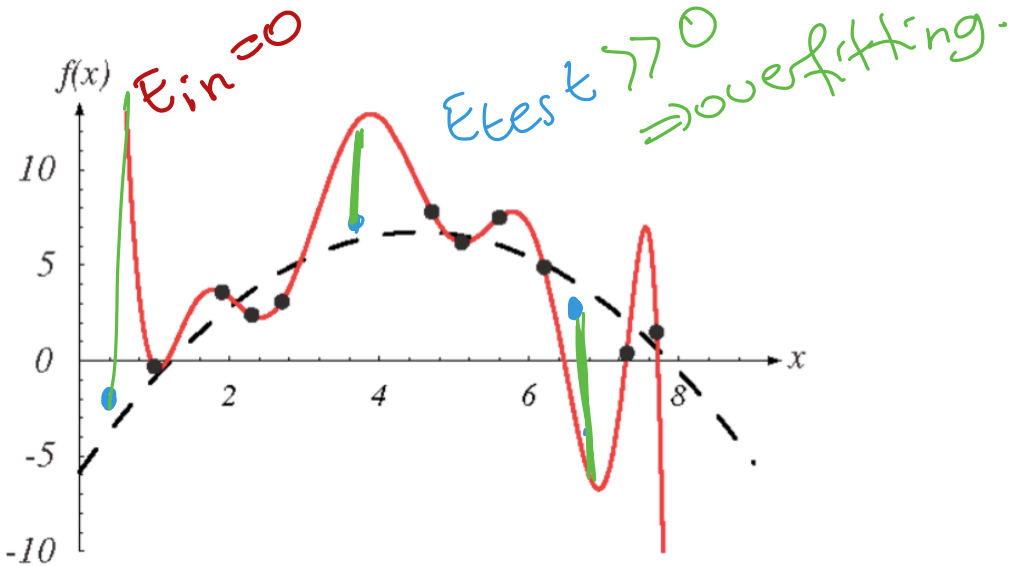


Figure 2: Problem of insufficient data is analogous to problems in curve fitting. The training data (black dots) are selected from a quadratic function plus Gaussian noise. A tenth-degree polynomial fits the data perfectly but we prefer a second-order polynomial for better generalization.

Problems of Dimensionality

- All of the commonly used classifiers can suffer from the curse of dimensionality.
- While an exact relationship between the probability of error, the number of training samples, the number of features, and the number of parameters is very difficult to establish, some guidelines have been suggested.
- It is generally accepted that using at least ten times as many training samples per class as the number of features ($n/d > 10$) is a good practice.
- The more complex the classifier, the larger should the ratio of sample size to dimensionality be.

Feature Reduction

- One way of coping with the problem of high dimensionality is to reduce the dimensionality by combining features.
- Issues in feature reduction:
  - Linear vs. non-linear transformations.
  - Use of class labels or not (depends on the availability of training data).
  - Training objective:
    - minimizing classification error (discriminative training),
    - minimizing reconstruction error (PCA),
    - maximizing class separability (LDA),
    - retaining interesting directions (projection pursuit),
    - making features as independent as possible (ICA),
    - embedding to lower dimensional manifolds (Isomap, LLE)

Feature Reduction

- Linear combinations are particularly attractive because they are simple to compute and are analytically tractable.
- Linear methods project the high-dimensional data onto a lower dimensional space.
- Advantages of these projections include
  - reduced complexity in estimation and classification,
  - ability to visually examine the multivariate data in two or three dimensions.

Feature Reduction

- Given $\mathbf{x} \in \mathbb{R}^d$, the goal is to find a linear transformation $\mathbf{A}$ that gives $\mathbf{y} = \mathbf{A}^T\mathbf{x} \in \mathbb{R}^{d'}$ where $d' < d$.
- Two classical approaches for finding optimal linear transformations are:
  - *Principal Components Analysis (PCA):* Seeks a projection that best represents the data in a least-squares sense.
  - *Linear Discriminant Analysis (LDA):* Seeks a projection that best separates the data in a least-squares sense.

Principal Component Analysis

- Given $\mathbf{x_1}, \ldots, \mathbf{x_n} \in \mathbb{R}^d$, the goal is to find a d'-dimensional subspace where the reconstruction error of $\mathbf{x_i}$ in this subspace is minimized.
- The criterion function for the reconstruction error can be defined in the least-squares sense as

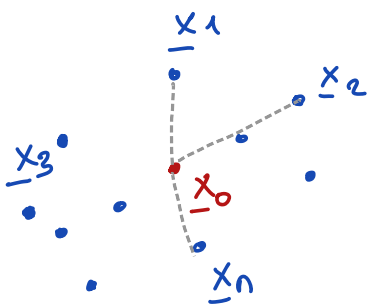
$$J_{d'} = \sum_{i=1}^{n} \left\| \sum_{k=1}^{d'} \mathbf{y}_{ik} \mathbf{e_k} - \mathbf{x_i} \right\|^2$$

where $\mathbf{e_1}, \ldots, \mathbf{e_{d'}}$ are the bases for the subspace (stored as the columns of $\mathbf{A}$) and $\mathbf{y_i}$ is the projection of $\mathbf{x_i}$ onto that subspace.

Principal Component Analysis

$D = \{\underline{x}_1, \dots, \underline{x}_n\}$ $\quad \underline{x}_i \in \mathbb{R}^d$

We want to "represent" all the vectors by a single vector $\underline{x}_0$.



$$J_0(\underline{x}_0) = \sum_{k=1}^{n} \|\underline{x}_0 - \underline{x}_k\|^2 \quad \text{(cost function)}$$

Find $\min_{\underline{x}_0} J_0(\underline{x}_0)$

$$\frac{\partial J_0}{\partial \underline{x}_0} = 0 \Rightarrow \boxed{\underline{x}_0 = \frac{1}{n}\sum_{k=1}^{n} \underline{x}_k}$$

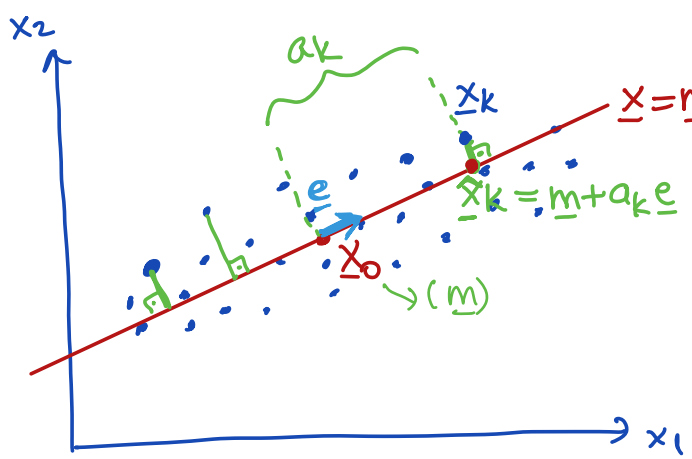
sample mean



(zero-dimensional representation of data set. It doesn't reveal any of the variability in the data)

Goal: we want to obtain a 1D representation of the data by projecting the data onto a line running through the sample mean.

$\underline{x} = [x_1, x_2]$



The eq^n of the red line

$$\underline{x} = \underline{m} + a\underline{e}$$

$\underline{m}$: $\underline{x}_0$: sample mean

$\underline{e}$: unit vector in the direction of the line

a : distance of any pt. $\underline{x}$ from the mean

Question: if we know $\underline{e}$ and $\underline{m}$, and we want to represent $\underline{x}_k$ by $\hat{\underline{x}}_k = \underline{m} + a_k \underline{e}$, how can we find a_k, the "optimal" set of coefficient?

$$J_1(a_1, a_2, \cdots, a_n; \underline{e}) = \sum_{k=1}^{n} \| \hat{\underline{x}}_k - \underline{x}_k \|^2$$

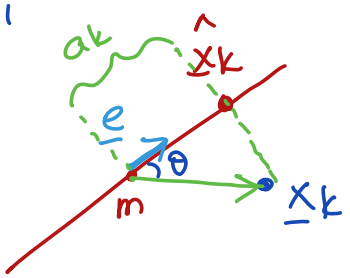
↳ projection of $\underline{x}_k$ onto the red line

$$J_1 = \sum_{k=1}^{n} \| (\underline{m} + a_k \underline{e}) - \underline{x}_k \|^2 \quad (1)$$

$$= \sum_{k=1}^{n} \| a_k \underline{e} - (\underline{x}_k - \underline{m}) \|^2$$

↳ $(a_k \underline{e} - (\underline{x}_k - \underline{m}))(a_k \underline{e}^t - (\underline{x}_k - \underline{m})^t)$

$$J_1 = \sum_{k=1}^{n} a_k^2 \|\underline{e}\|^2 - 2\sum_{k=1}^{n} a_k \underbrace{\underline{e}^t (\underline{x}_k - \underline{m})}_{a_k} + \sum_{k=1}^{n} \|\underline{x}_k - \underline{m}\|^2 \quad (2)$$



Let $\|\underline{e}\| = 1$

$$\frac{\partial J_1}{\partial a_k} = 0$$

$$\Rightarrow 2a_k - 2\underline{e}^t (\underline{x}_k - \underline{m}) = 0$$

$$a_k = \underline{e}^t(\underline{x}_k - \underline{m}) \quad (3)$$

To obtain a least-squares solⁿ project the vector $\underline{x}_k$ onto the line in the direction of $\underline{e}$ and passing through $\underline{m}$.

Question: How can we find the "best" direction $\underline{e}$?

Define scatter matrix $\underline{\underline{S}}$

$$\underline{\underline{S}} = \sum_{k=1}^{n} (\underline{x}_k - \underline{m})(\underline{x}_k - \underline{m})^t$$

($n-1$ times the sample covariance matrix)

Substitute (3) into (2)

$$J_1(\underline{e}) = \sum_{k=1}^{n} a_k^2 - 2\sum_{k=1}^{n} a_k^2 + \sum_{k=1}^{n} \|\underline{x}_k - \underline{m}\|^2$$

put $a_k = \underline{e}^T(\underline{x}_k - \underline{m})$

$$= -\sum_{k=1}^{n} \left[\underline{e}^T(\underline{x}_k - \underline{m})\right]^2 + \sum_{k=1}^{n} \|\underline{x}_k - \underline{m}\|^2$$

$\underline{e}^T(\underline{x}_k - \underline{m})$: a scalar

$b^2 = bb^t$

$$= -\sum_{k=1}^{n} \left[\underline{e}^t(\underline{x}_k - \underline{m})(\underline{x}_k - \underline{m})^t \underline{e}\right] + \sum_{k=1}^{n} \|\underline{x}_k - \underline{m}\|^2$$

$(\underline{x}_k - \underline{m})(\underline{x}_k - \underline{m})^t = \underline{\underline{S}}$

$$J_1(\underline{e}) = -\underline{e}^t \underline{\underline{S}}\, \underline{e} + \sum_{k=1}^{n} \|\underline{x}_k - \underline{m}\|^2$$

ignore since doesn't depend on $\underline{e}$

maximize $\underline{e}^t \underline{\underline{S}}\, \underline{e}$ subject to the constraint $\|\underline{e}\| = 1$

using Lagrange multipliers:

$$u = \underline{e}^t \underline{\underline{S}}\, \underline{e} - \lambda (\underline{e}^t \underline{e} - 1)$$

$$\frac{\partial u}{\partial \underline{e}} = 2\underline{\underline{S}}\, \underline{e} - 2\lambda \underline{e} = 0$$

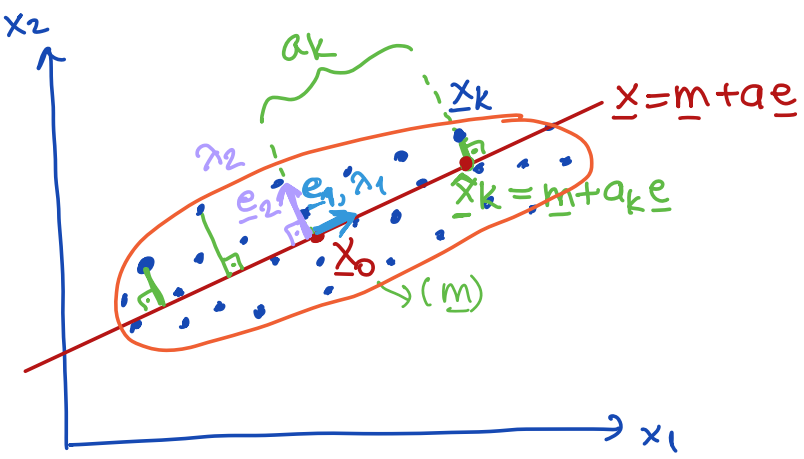
$$\boxed{\underline{\underline{S}}\, \underline{e} = \lambda \underline{e}}$$

($\underline{e}$ must be an eigenvector of the scatter matrix)

λ : eigenvalue

maximize $\underline{e}^t \underline{\underline{S}}\, \underline{e} = \lambda \underbrace{\underline{e}^t \underline{e}}_{1} = \lambda$

$\Rightarrow$ to maximize $\underline{e}^t \underline{\underline{S}}\, \underline{e}$, select the eigenvector corresponding to the largest eigenvalue of the scatter matrix $\underline{\underline{S}}$.



if $\lambda_1 > \lambda_2$ choose $\underline{e}_1$

else choose $\underline{e}_2$

Result: To find the "best" 1D projection of the data (in the least sum-of-squares sense), project the data onto a line through sample mean, in the direction of the eigenvector of the scatter matrix having the largest eigenvalue.

Extend from 1D projection to d' directional projection.

$$\hat{\underline{x}} = \underline{m} + \sum_{i=1}^{d'} a_i \underline{e}_i \quad \text{where } d' \leq d$$

(d: dimension of data)

criterion function:

$$J_{d'} = \sum_{k=1}^{n} \| \hat{\underline{x}}_k - \underline{x}_k \|^2$$

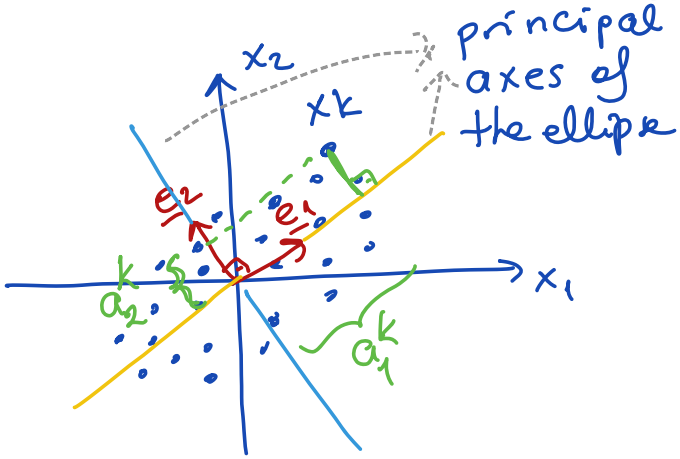
$\hat{\underline{x}}_k$: projection of $\underline{x}_k$ onto the d' dimensional space

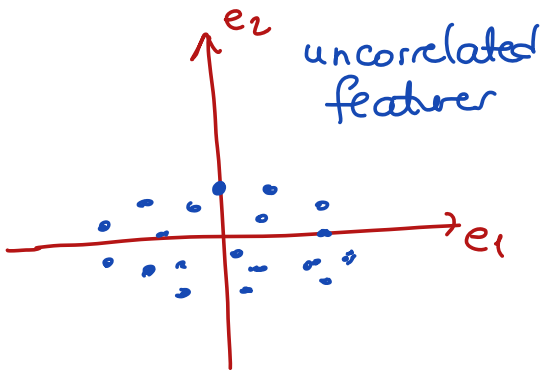
$$J_{d'} = \sum_{k=1}^{n} \| (\underline{m} + \sum_{i=1}^{d'} a_i^k e_i) - \underline{x}_k \|^2$$

is minimized when the vectors $\underline{e}_1, \cdots \underline{e}_{d'}$ are eigenvectors of the scatter matrix having the largest eigenvalues.

- Since scatter matrix is real and symmetric the eigenvectors are orthogonal.
- They form a natural set of basis vectors for representing any feature vector $\underline{x}$.

⊛ The coefficients a_i are the components of $\underline{x}$ in that basis, called as principal components.





if $d' = d \Rightarrow$ we have a change of basis problem.

put origin to $\underline{m}$

axis 1 // $\underline{e}_1$

axis 2 // $\underline{e}_2$

Principal Component Analysis (PCA) Summary

Given a dataset of n vectors in a d-dimensional space

$\underline{x}_1, \underline{x}_2, \cdots \underline{x}_n$

1) Find the sample mean $\underline{m} = \frac{1}{n} \sum_{k=1}^{n} \underline{x}_k$

2) Find the scatter matrix

$$\underline{\underline{S}} = \sum_{k=1}^{n} (\underline{x}_k - \underline{m})(\underline{x}_k - \underline{m})^t \,, \quad \underline{\underline{S}}_{d \times d}$$

3) Find the eigenvectors and eigenvalues of $\underline{\underline{S}}$, and order them in decreasing order w.r.t the eigenvalue. $\{\underline{e}_1, \underline{e}_2, \cdots, \underline{e}_d\}$

$\{\lambda_1, \lambda_2, \cdots, \lambda_d\} \rightarrow$ decreasing order

4) Form the $\underline{\underline{W}}$ matrix

$\mathbb{R}^d$

$$\underset{d \times d'}{\underline{\underline{W}}} = \left[\underline{e}_1 \,\vdots\, \underline{e}_2 \,\vdots\, \cdots \,\vdots\, \underline{e}_{d'} \right]$$

$$\underbrace{\underline{a}_k}_{d' \times 1} = \underbrace{\underline{\underline{W}}^t}_{d' \times d} \underbrace{(\underline{x}_k - \underline{m})}_{d \times 1}$$

Project vector $\underline{x}_k$ onto the d' dimensional space, where the d' columns of $\underline{\underline{W}}$ are the leading d' eigenvectors of $\underline{\underline{S}}$. $\underline{a}_k \in \mathbb{R}^{d'}$

- After this linear transformation, we get a d' dimensional space whose dimensions are the eigenvectors, and the variances over new dimensions are the eigenvalues.

5) How to choose d'?

The proportion of variance explained by d' principal components:

$$\frac{\lambda_1 + \lambda_2 + \cdots \lambda_{d'}}{\lambda_1 + \lambda_2 + \cdots + \lambda_{d'} + \cdots \lambda_d} = 0.9 \left[\begin{array}{l} \text{keeping 90\% of} \\ \text{total variance} \end{array} \right]$$

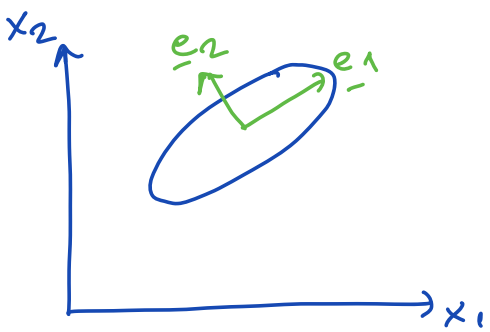
total variance in $\mathbb{R}^d$.

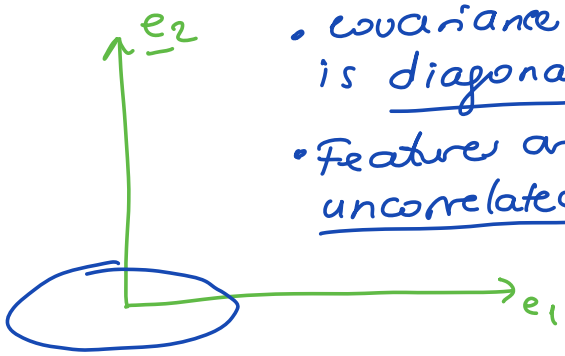
$$\underline{x}_k = \begin{bmatrix} x_{1k} \\ \vdots \\ x_{dk} \end{bmatrix}$$

- If the dimensions are highly correlated there will be a small # of eigenvectors with large eigenvalues and d' will be much smaller than d.

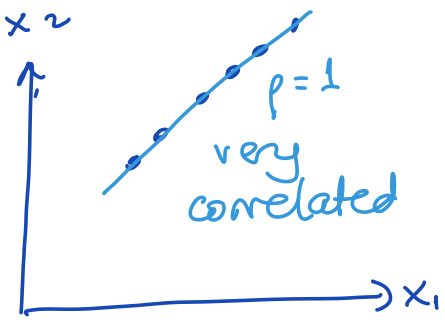
- If the dimensions are not correlated there will be a large # of eigenvectors with large eigenvalues and $d' \approx d$. $\Rightarrow$ There will be no gain of PCA.

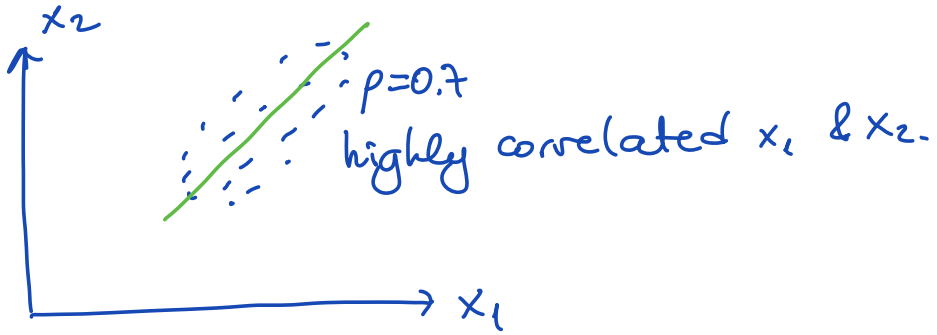
PCA centers the sample and rotates the axis to line up with the directions of highest variance.

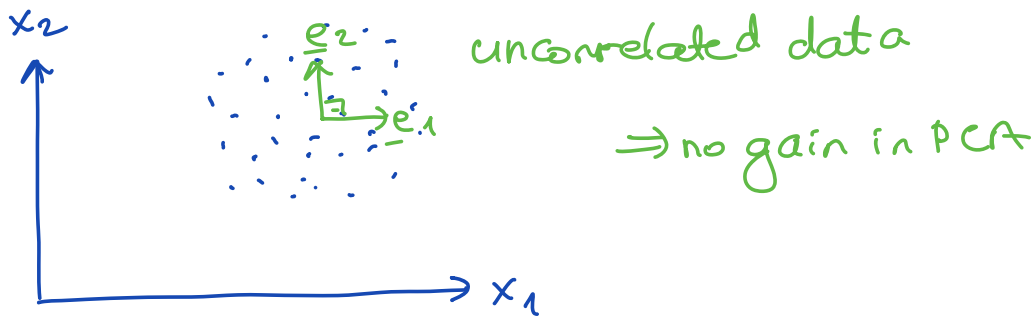




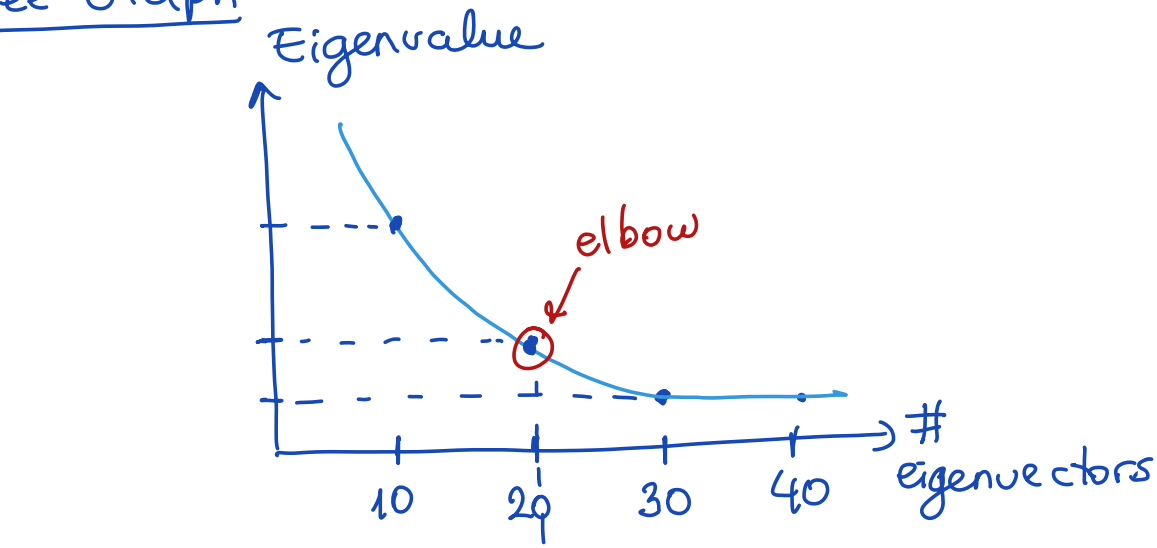
- covariance matrix is diagonalized
- Features are uncorrelated

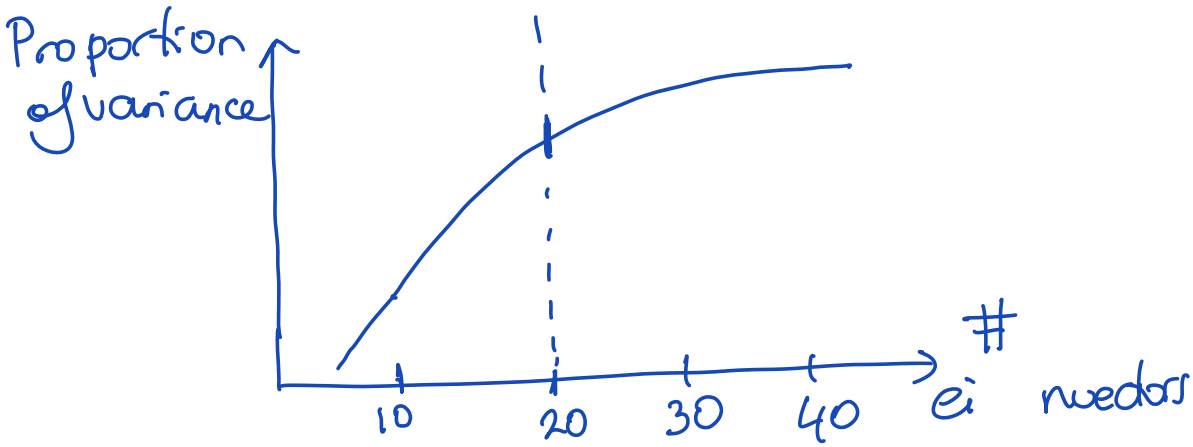



x_2
$\rho = 0.7$
highly correlated x_1 & x_2.
x_1


x_2
e_2
e_1
uncorrelated data
⇒ no gain in PCA
x_1

Scree Graph :


Eigenvalue
elbow
#
eigenvectors
10
20
30
40


Proportion
of variance
#
eigenvectors
10
20
30
40

At the "elbow" adding another eigenvector does not significantly increase the variance explained.

Note: $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ $x_1 \to (0,1)$, $x_2 \to (-10^3, 10^3)$ } different ranges

Before using PCA preprocess the data so that each dimension has mean 0 and unit variance. This is needed since scales of each dimension may be different.

$$\frac{x_1 - m_{x_1}}{\sigma_{x_1}}$$

m_{x_1}, σ_{x_1} ← keep and use the same values on test data.

ge

Principal Component Analysis

- It can be shown that $J_{d'}$ is minimized when $\mathbf{e_1}, \ldots, \mathbf{e}_{d'}$ are the d' eigenvectors of the *scatter matrix*

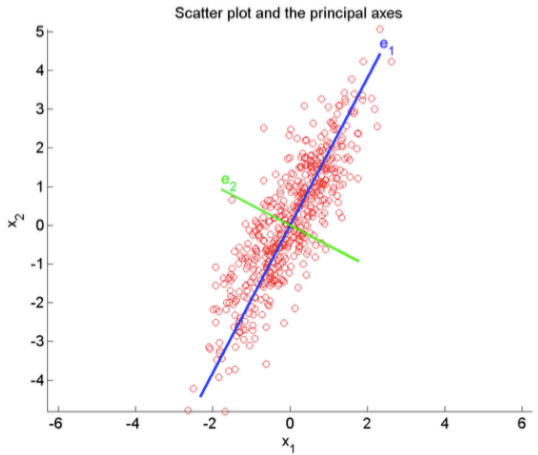
$$S = \sum_{i=1}^{n} (\mathbf{x_i} - \boldsymbol{\mu})(\mathbf{x_i} - \boldsymbol{\mu})^T$$

  having the largest eigenvalues.
- The coefficients $\mathbf{y} = (\mathbf{y}_i, \ldots, \mathbf{y}_{d'})^T$ are called the *principal components*.
- When the eigenvectors are sorted in descending order of the corresponding eigenvalues, the greatest variance of the data lies on the first principal component, the second greatest variance on the second component, etc.

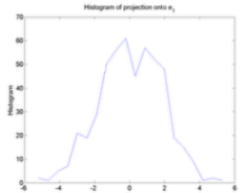
Principal Component Analysis

- Often there will be just a few large eigenvalues, and this implies that the d'-dimensional subspace contains the signal and the remaining $d - d'$ dimensions generally contain noise.
- The actual subspace where the data may lie is related to the *intrinsic dimensionality* that determines whether the given d-dimensional patterns can be described adequately in a subspace of dimensionality less than d.
- The geometric interpretation of intrinsic dimensionality is that the entire data set lies on a topological d'-dimensional hypersurface.
- Note that the intrinsic dimensionality is not the same as the linear dimensionality which is related to the number of significant eigenvalues of the scatter matrix of the data.

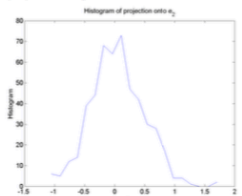
Examples



(a) Scatter plot.



(b) Projection onto e_1.



(c) Projection onto e_2.

Figure 3: Scatter plot (red dots) and the principal axes for a bivariate sample. The blue line shows the axis e_1 with the greatest variance and the green line shows the axis e_2 with the smallest variance. Features are now uncorrelated.

Examples

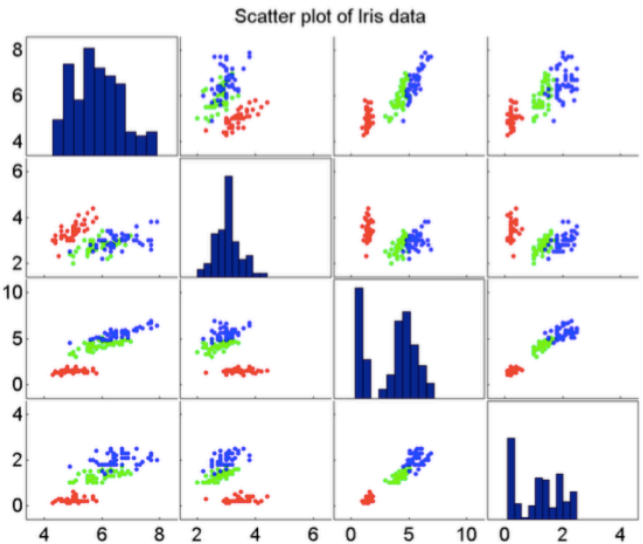


Figure 4: Scatter plot of the iris data. Diagonal cells show the histogram for each feature. Other cells show scatters of pairs of features x_1, x_2, x_3, x_4 in top-down and left-right order. Red, green and blue points represent samples for the setosa, versicolor and virginica classes, respectively.

Examples

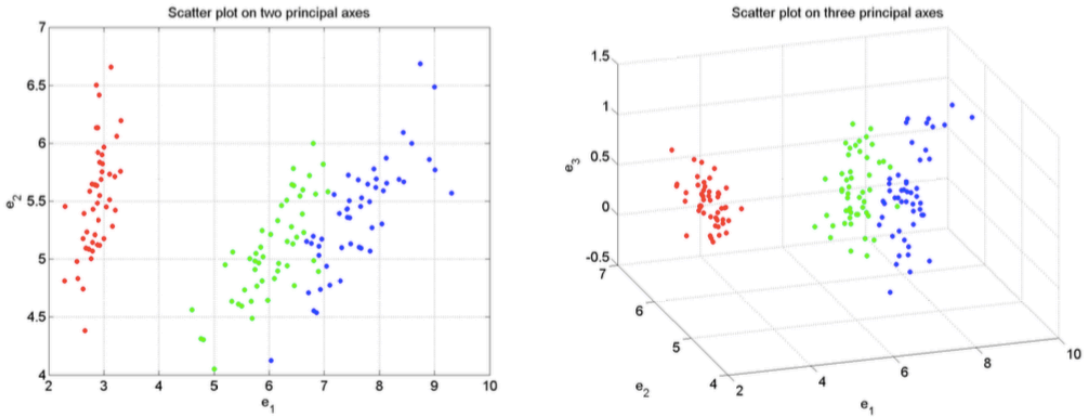
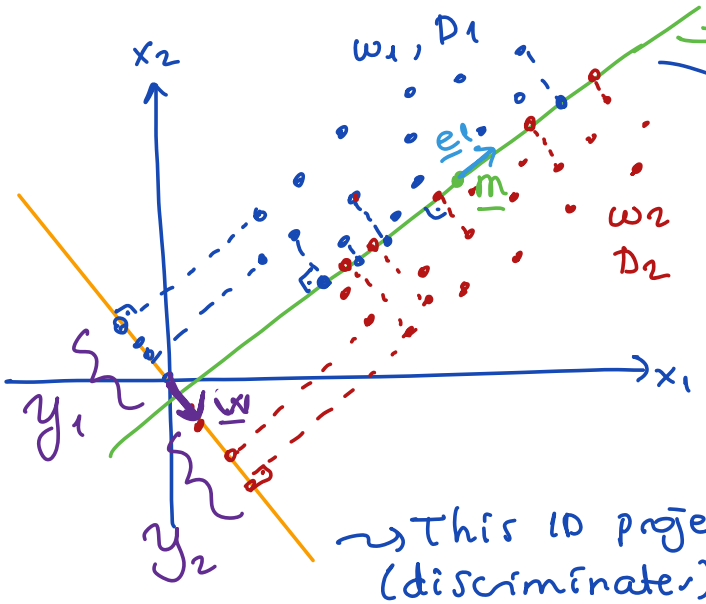


Figure 5: Scatter plot of the projection of the iris data onto the first two and the first three principal axes. Red, green and blue points represent samples for the setosa, versicolor and virginica classes, respectively.

Fisher Linear Discriminant (Linear Discriminant Analysis: LDA)
(FLD)



→ This 1D projection using PCA of the whole data does not seperate classes well.

→ This 1D projection seperates (discriminates) the two classes well.

PCA : goal is best representation of data ⇏ best discrimination

Discriminant analysis : seek directions that are efficient in discriminating btwn classes.

Problem : Project data from d dimensions onto a line (direction $\underline{w}$) s.t. the projected samples are "well-seperated".

Suppose we have n d-dimensional samples:

$$\underline{x}_1, \underline{x}_2, \dots, \underline{x}_n$$

n_1 of them are in subset D_1 labeled ω_1.

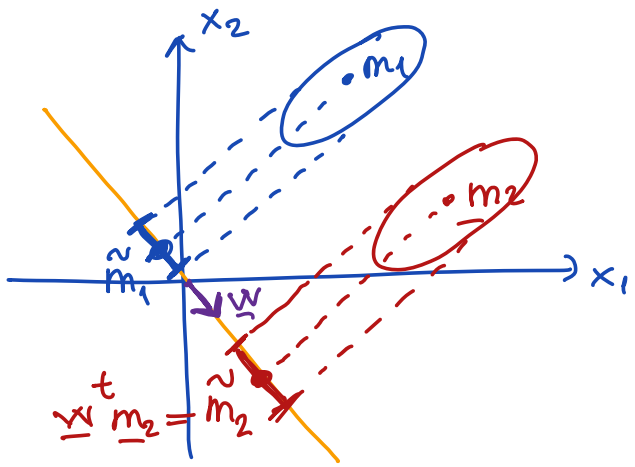
n_2 " " D_2 " ω_2.

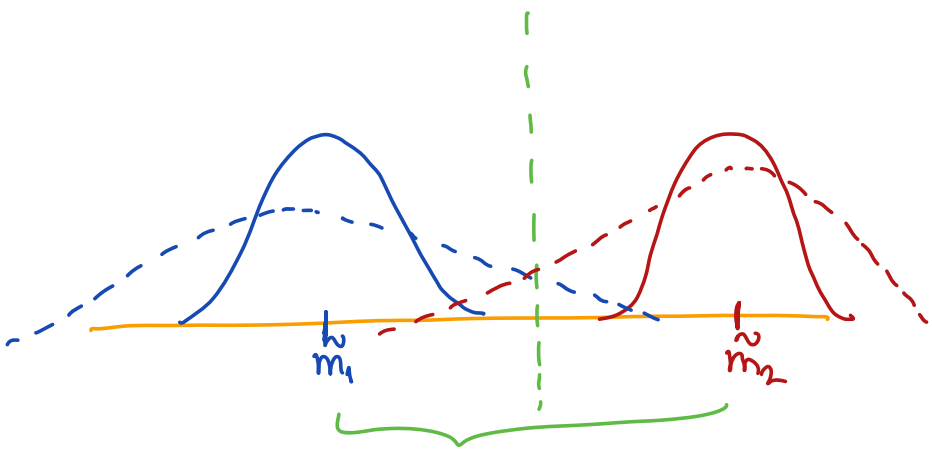
Form a linear combination of the components of $\underline{x}$.

$$y = \underline{w}^t \underline{x}$$

obtain corresponding samples $y_1, y_2, \ldots, y_n$ divided into two subsets Y_1, Y_2.

- If $\|\underline{w}\| = 1$, each y_i is the projection of x_i (scale is not important)
- want the projections to be <u>well-separated</u>.
- A <u>measure of separation</u> btwn. projected pts:





maximize distance btwn. projected means.

$$\underline{m}_i = \frac{1}{n_i} \sum_{\underline{x} \in D_i} \underline{x}$$

Sample mean of projected ptrs:

$$\tilde{m}_i = \frac{1}{n_i} \sum_{y \in Y_i} y = \frac{1}{n_i} \sum_{\underline{x} \in D_i} \underline{w}^t \underline{x} = \underbrace{\underline{w}^t \underline{m}_i}_{\text{projection of } \underline{m}_i}$$

Distance btwn projected means:

$$|\tilde{m}_1 - \tilde{m}_2| = |\underline{w}^t (\underline{m}_1 - \underline{m}_2)|$$

→ want this to be large for good class seperation

want it large relative to some measure of standard deviations:

Scatter for projected samples:

$$\tilde{s}_i^2 = \sum_{y_i \in Y_i} (y_i - \tilde{m}_i)^2$$

$\frac{1}{n}(\tilde{s}_1^2 + \tilde{s}_2^2)$ is estimate of the variance of pooled data.

total within class scatter of projected samples.

Fisher Linear Discriminant maximizes the criterion function:

$$J(\underline{w}) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

→ maximize distance btwn means

→ to minimize variance

Scatter matrices

$$\underline{\underline{S}}_i = \sum_{x \in D_i} (\underline{x} - \underline{m}_i)(\underline{x} - \underline{m}_i)^t$$

and $\underline{\underline{S}}_W = \underline{\underline{S}}_1 + \underline{\underline{S}}_2$ (within class scatter matrix)
symmetric, positive definite
usually nonsingular if $n > d$

$$\tilde{s}_i^2 = \sum_{x \in D_i} (\underbrace{\underline{w}^t \underline{x}}_{y} - \underbrace{\underline{w}^t \underline{m}_i}_{\tilde{m}_i})^2$$

scalar: $a^2 = aa^t$

$$(\underline{w}^t \underline{x} - \underline{w}^t \underline{m}_i)(\underline{w}^t \underline{x} - \underline{w}^t \underline{m}_i)^t$$

$$= \sum_{x \in D_i} \underline{w}^t (\underline{x} - \underline{m}_i)(\underline{x} - \underline{m}_i)^t \underline{w} = \underline{w}^t \underline{\underline{S}}_i \underline{w}$$

$$\tilde{s}_1^2 + \tilde{s}_2^2 = \underline{w}^t \underline{\underline{S}}_1 \underline{w} + \underline{w}^t \underline{\underline{S}}_2 \underline{w} = \underline{w}^t \underline{\underline{S}}_W \underline{w}$$

$$(\tilde{m}_1 - \tilde{m}_2)^2 = (\underline{w}^t \underline{m}_1 - \underline{w}^t \underline{m}_2)^2 = \underline{w}^t \underbrace{(\underline{m}_1 - \underline{m}_2)(\underline{m}_1 - \underline{m}_2)^t}_{\underline{\underline{S}}_B} \underline{w}$$

$\underline{\underline{S}}_B$: between class scatter matrix
- symmetric

• rank is at most one

$$(\tilde{m}_1 - \tilde{m}_2)^2 = \underline{w}^t \underline{\underline{S}}_B \underline{w}$$

$$J(\underline{w}) = \frac{\underline{w}^t \underline{\underline{S}}_B \underline{w}}{\underline{w}^t \underline{\underline{S}}_W \underline{w}}$$

maximize $J(\underline{w})$ w.r.t $\underline{w}$, s.t. $\|\underline{w}\| = 1$

$\Rightarrow \underline{\underline{S}}_B \underline{w} = \lambda \underline{\underline{S}}_W \underline{w}$, λ constant

(easy to show using Lagrange multipliers)

if $\underline{\underline{S}}_W$ is invertible:

$$\underline{\underline{S}}_W^{-1} \underline{\underline{S}}_B \underline{w} = \lambda \underline{w}$$

$$\underline{\underline{S}}_B \underline{w} = (\underline{m}_1 - \underline{m}_2)(\underline{m}_1 - \underline{m}_2)^t \underline{w}$$

$$= \text{scalar} \cdot (\underline{m}_1 - \underline{m}_2)$$

$\Rightarrow$ the solution

$$\underline{w} = \underline{\underline{S}}_W^{-1} (\underline{m}_1 - \underline{m}_2)$$

↓

Fisher's linear discriminant
(canonical variate)

Multiple Discriminant Analysis

* c-class problem
* LDA involves c-1 discriminant functions
* Thus, the projection is from a d-dimensional

space to a (max.) (c-1) dimensional space, if $d \geq c$.

Generalization

$$\underline{\underline{S}}_W = \sum_{i=1}^{c} \underline{\underline{S}}_i$$

Total mean vector $\underline{m} = \frac{1}{n} \sum_{\underline{x}} \underline{x}$

" scatter matrix $\underline{\underline{S}}_T = \sum_{\underline{x}} (\underline{x} - \underline{m})(\underline{x} - \underline{m})^t$

steps in book

$$\boxed{\underline{\underline{S}}_T = \underline{\underline{S}}_W + \underline{\underline{S}}_B}$$

$$\underline{\underline{S}}_B = \sum_{i=1}^{c} n_i (\underline{m}_i - \underline{m})(\underline{m}_i - \underline{m})^t$$

rank is at most (c-1)

Cost function

determinant

$$\max_{\underline{\underline{W}}} J(\underline{\underline{W}}) = \frac{|\tilde{\underline{\underline{S}}}_B|}{|\tilde{\underline{\underline{S}}}_W|} = \frac{|\underline{\underline{W}}^t \underline{\underline{S}}_B \underline{\underline{W}}|}{|\underline{\underline{W}}^t \underline{\underline{S}}_W \underline{\underline{W}}|}$$

→ represents btwn class scatter

→ represents within class scatter

$\underline{\underline{W}}$ ← matrix

$\underline{\underline{W}}$: columns are the eigenvectors corresponding to the largest eigenvalues in :

$$\underline{\underline{S}}_B \underline{w}_i = \lambda_i \underline{\underline{S}}_W \underline{w}_i$$

$$\underline{\underline{S}}_W^{-1} \underline{\underline{S}}_B \underline{w}_i = \lambda_i \underline{w}_i$$

find the eigenvectors (c-1) nonzero eigenvectors

Linear Discriminant Analysis

- Whereas PCA seeks directions that are efficient for representation, discriminant analysis seeks directions that are efficient for discrimination.
- Given $\mathbf{x_1}, \ldots, \mathbf{x_n} \in \mathbb{R}^d$ divided into two subsets $\mathcal{D}_1$ and $\mathcal{D}_2$ corresponding to the classes w_1 and w_2, respectively, the goal is to find a projection onto a line defined as

$$y = \mathbf{w}^T \mathbf{x}$$

where the points corresponding to $\mathcal{D}_1$ and $\mathcal{D}_2$ are well separated.

Linear Discriminant Analysis

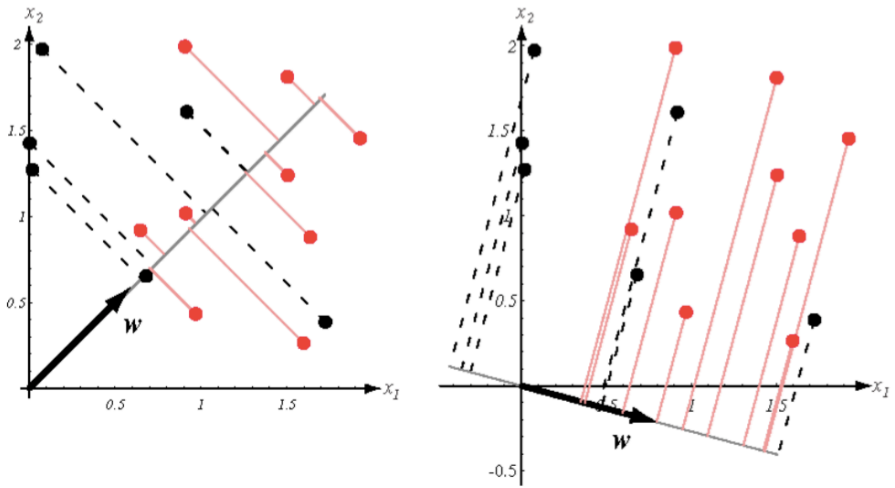


Figure 6: Projection of the same set of samples onto two different lines in the directions marked as $\mathbf{w}$. The figure on the right shows greater separation between the red and black projected points.

Linear Discriminant Analysis

- The criterion function for the best separation can be defined as

$$J(\mathbf{w}) = \frac{|\tilde{m}_1 - \tilde{m}_2|^2}{\tilde{s}_1^2 + \tilde{s}_2^2}$$

  where $\tilde{m}_i = \frac{1}{\#\mathcal{D}_i} \sum_{y \in w_i} y$ is the sample mean and $\tilde{s}_i^2 = \sum_{y \in w_i} (y - \tilde{m}_i)^2$ is the scatter for the projected samples labeled w_i.

- This is called the *Fisher's linear discriminant* with the geometric interpretation that the best projection makes the difference between the means as large as possible relative to the variance.

Linear Discriminant Analysis

- To compute the optimal $\mathbf{w}$, we define the *scatter matrices* $\mathbf{S}_i$

$$\mathbf{S}_i = \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T \text{ where } \mathbf{m}_i = \frac{1}{\#\mathcal{D}_i} \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{x},$$

the *within-class scatter matrix* $\mathbf{S_W}$

$$\mathbf{S_W} = \mathbf{S_1} + \mathbf{S_2},$$

and the *between-class scatter matrix* $\mathbf{S_B}$

$$\mathbf{S_B} = (\mathbf{m_1} - \mathbf{m_2})(\mathbf{m_1} - \mathbf{m_2})^T.$$

Linear Discriminant Analysis

- Then, the criterion function becomes

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S_B} \mathbf{w}}{\mathbf{w}^T \mathbf{S_W} \mathbf{w}}$$

and the optimal $\mathbf{w}$ can be computed as

$$\mathbf{w} = \mathbf{S_W^{-1}}(\mathbf{m_1} - \mathbf{m_2}).$$

- Note that, $\mathbf{S_W}$ is symmetric and positive semidefinite, and it is usually nonsingular if $n > d$. $\mathbf{S_B}$ is also symmetric and positive semidefinite, but its rank is at most 1.

Linear Discriminant Analysis

- Generalization to c classes involves $c-1$ discriminant functions where the projection is from a d-dimensional space to a $(c-1)$-dimensional space ($d \geq c$).
- The scatter matrices $\mathbf{S}_i$ are computed as

$$\mathbf{S}_i = \sum_{\mathbf{x} \in \mathcal{D}_i} (\mathbf{x} - \mathbf{m}_i)(\mathbf{x} - \mathbf{m}_i)^T \text{ where } \mathbf{m}_i = \frac{1}{\#\mathcal{D}_i} \sum_{\mathbf{x} \in \mathcal{D}_i} \mathbf{x}.$$

- The within-class scatter matrix $\mathbf{S}_\mathbf{W}$ is computed as

$$\mathbf{S}_\mathbf{W} = \sum_{i=1}^{c} \mathbf{S}_i.$$

Linear Discriminant Analysis

- The between-class scatter matrix $\mathbf{S_B}$ is computed as

$$\mathbf{S_B} = \sum_{i=1}^{c} (\#\mathcal{D}_i)(\mathbf{m_i} - \mathbf{m})(\mathbf{m_i} - \mathbf{m})^T$$

  where $\mathbf{m} = \frac{1}{n} \sum_{\mathbf{x}} \mathbf{x}$ is the total mean vector.

- Then, the criterion function becomes

$$J(\mathbf{W}) = \frac{|\mathbf{W}^T \mathbf{S_B} \mathbf{W}|}{|\mathbf{W}^T \mathbf{S_W} \mathbf{W}|}$$

  where $\mathbf{W}$ is the d-by-$(c-1)$ transformation matrix and $|\cdot|$ represents the determinant.

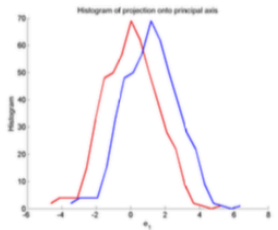
Linear Discriminant Analysis

- It can be shown that $J(\mathbf{W})$ is maximized when the columns of $\mathbf{W}$ are the eigenvectors of $\mathbf{S}_\mathbf{W}^{-1}\mathbf{S}_\mathbf{B}$ having the largest eigenvalues.
- Because $\mathbf{S}_\mathbf{B}$ is the sum of c matrices of rank one or less, and because only $c-1$ of these are independent, $\mathbf{S}_\mathbf{B}$ is of rank $c-1$ or less. Thus, no more than $c-1$ of the eigenvalues are nonzero.
- Once the transformation from the d-dimensional original feature space to a lower dimensional subspace is done using PCA or LDA, parametric or non-parametric methods can be used to train Bayesian classifiers.

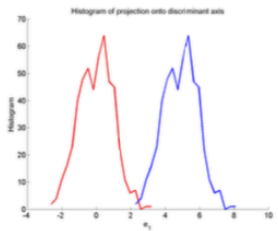
Examples



(a) Scatter plot.



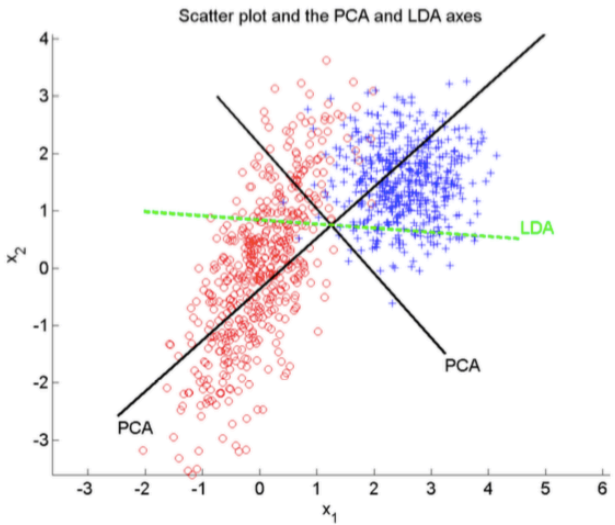
(b) Projection onto the first PCA axis.



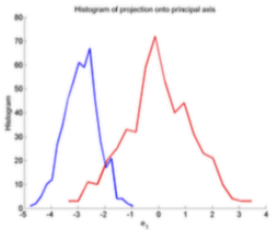
(c) Projection onto the first LDA axis.

Figure 7: Scatter plot and the PCA and LDA axes for a bivariate sample with two classes. Histogram of the projection onto the first LDA axis shows better separation than the projection onto the first PCA axis.

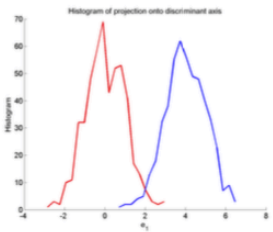
Examples



(a) Scatter plot.



(b) Projection onto the first PCA axis.



(c) Projection onto the first LDA axis.

Figure 8: Scatter plot and the PCA and LDA axes for a bivariate sample with two classes. Histogram of the projection onto the first LDA axis shows better separation than the projection onto the first PCA axis.

Examples

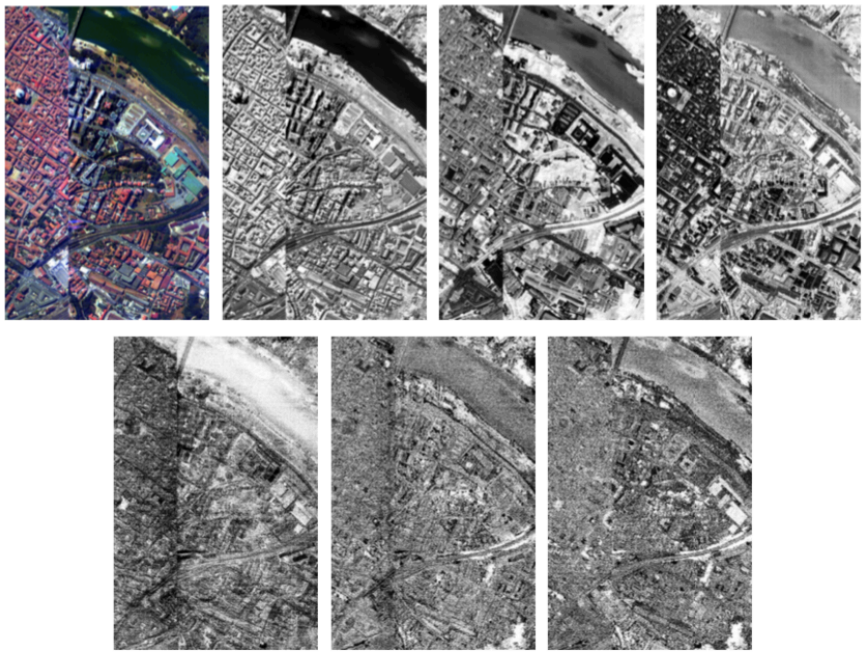


Figure 11: A satellite image and the first six PCA bands (after projection) Histogram equalization was applied to all images for better visualization.

Examples

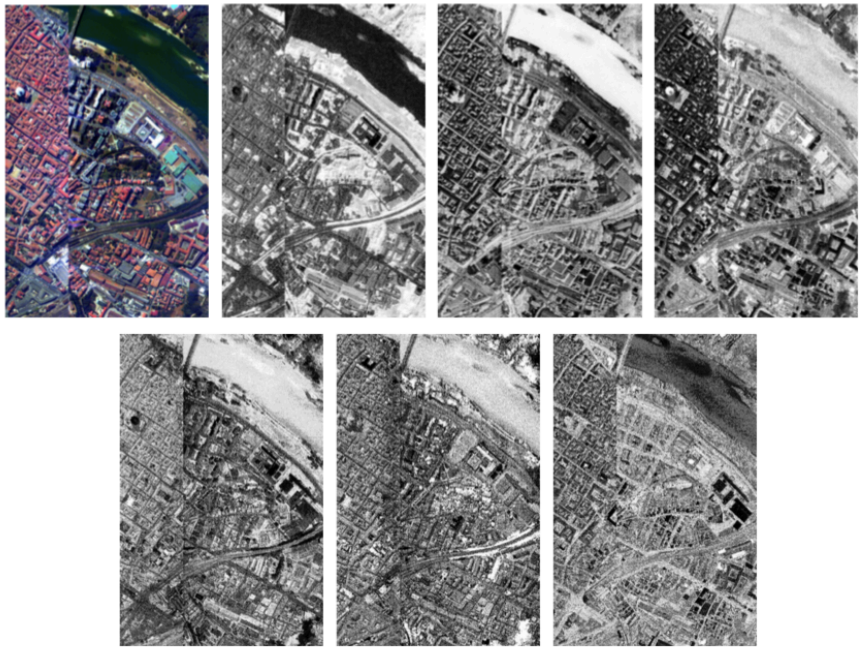


Figure 12: A satellite image and the six LDA bands (after projection). Histogram equalization was applied to all images for better visualization.

Examples



Figure 13: Example face images. (Taken from http://www.geop.ubc.ca/CDSST/eigenfaces.html.)

Examples

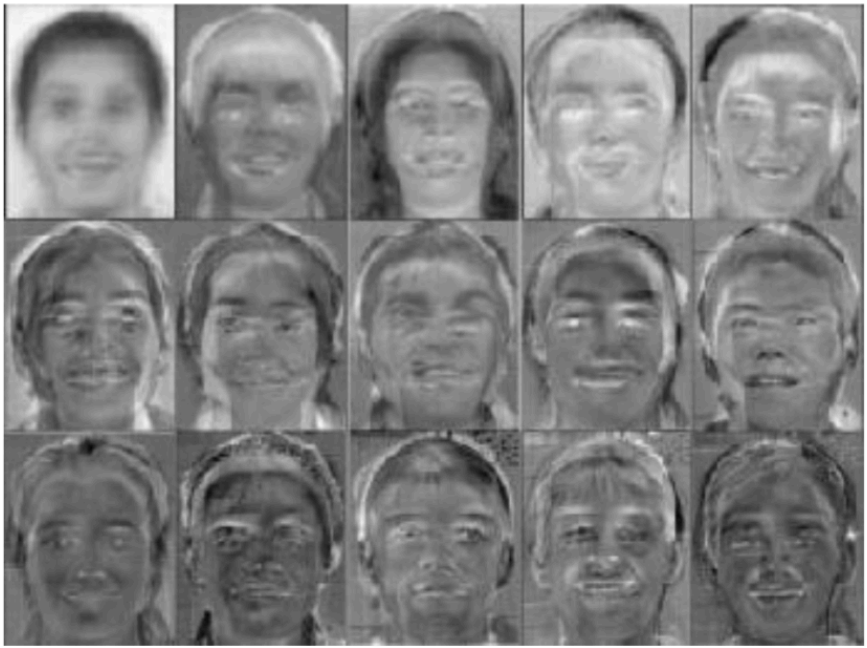


Figure 14: Eigenvectors (principal axes) of the face images (often referred to as eigenfaces).

Feature Selection

- An alternative to feature reduction that uses linear or non-linear combinations of features is feature selection that reduces dimensionality by selecting subsets of existing features.
- The first step in feature selection is to define a criterion function that is often a function of the classification error.
- Note that, the use of classification error in the criterion function makes feature selection procedures dependent on the specific classifier used.

Feature Selection

- The most straightforward approach would require
  - examining all $\binom{d}{m}$ possible subsets of size m,
  - selecting the subset that performs the best according to the criterion function.
- The number of subsets grows combinatorially, making the exhaustive search impractical.
- Iterative procedures are often used but they cannot guarantee the selection of the optimal subset.

Feature Selection

- *Sequential forward selection:*
  - First, the best single feature is selected.
  - Then, pairs of features are formed using one of the remaining features and this best feature, and the best pair is selected.
  - Next, triplets of features are formed using one of the remaining features and these two best features, and the best triplet is selected.
  - This procedure continues until all or a predefined number of features are selected.

Feature Selection

- *Sequential backward selection:*
  - First, the criterion function is computed for all d features.
  - Then, each feature is deleted one at a time, the criterion function is computed for all subsets with $d-1$ features, and the worst feature is discarded.
  - Next, each feature among the remaining $d-1$ is deleted one at a time, and the worst feature is discarded to form a subset with $d-2$ features.
  - This procedure continues until one feature or a predefined number of features are left.

Examples

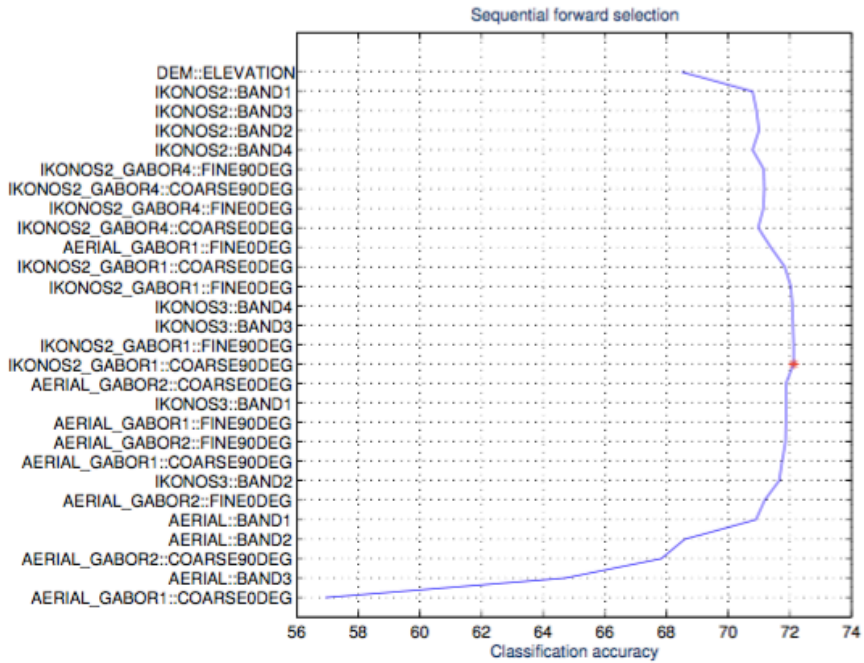


Figure 24: Results of sequential forward feature selection for classification of a satellite image using 28 features. x-axis shows the classification accuracy (%) and y-axis shows the features added at each iteration (the first iteration is at the bottom). The highest accuracy value is shown with a star.

Examples

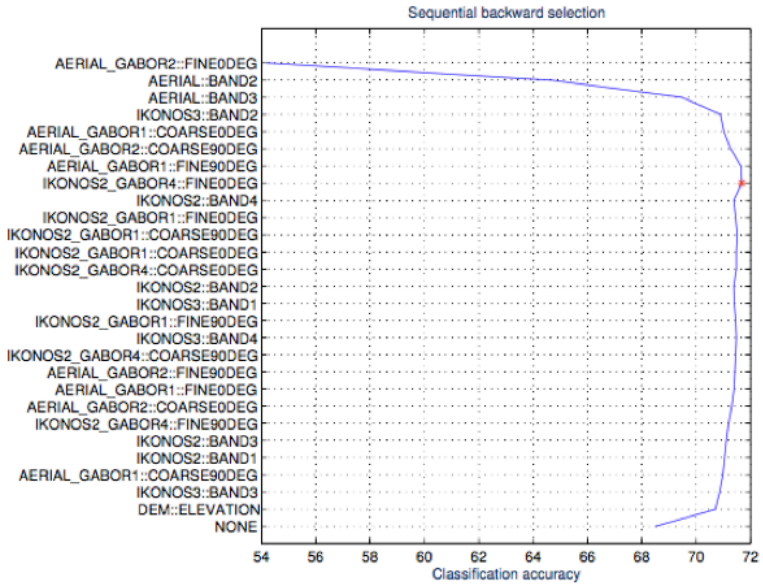


Figure 25: Results of sequential backward feature selection for classification of a satellite image using 28 features. x-axis shows the classification accuracy (%) and y-axis shows the features removed at each iteration (the first iteration is at the bottom). The highest accuracy value is shown with a star.

Summary

- The choice between feature reduction and feature selection depends on the application domain and the specific training data.
- Feature selection leads to savings in computational costs and the selected features retain their original physical interpretation.
- Feature reduction with transformations may provide a better discriminative ability but these new features may not have a clear physical meaning.